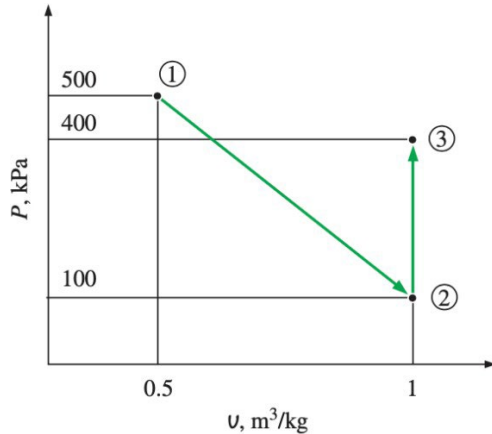


Tutorial Questions Week 4

Moving Boundary Work

1) Calculate the total work, in kJ, for process 1–3 shown in the figure below when the system consists of 3 kg of nitrogen.



2) 2 m^3 of saturated liquid water at 220°C is expanded isothermally in a closed system until its quality is 80 percent. Determine the total work produced by this expansion, in kJ.

3) The volume of 1 kg of helium in a piston–cylinder device is initially 4 m^3 . Now helium is compressed to 1 m^3 while its pressure is maintained constant at 120 kPa. Determine the initial and final temperatures of helium as well as the work required to compress it, in kJ.

4) A gas is compressed from an initial volume of 0.2 m^3 to a final volume of 0.1 m^3 . During the quasi-equilibrium process, the pressure changes with volume according to the relation $P = aV + b$, where $a = -1500 \text{ kPa/m}^3$ and $b = 400 \text{ kPa}$. Calculate the work done during this process: (a) by plotting the process on a P-V diagram and finding the area under the process curve and (b) by performing the necessary integrations.

Close Energy System Analysis

5) A 0.8-m^3 rigid tank contains refrigerant-134a initially at 180 kPa and 45 percent quality. Heat is now transferred to the refrigerant until the pressure reaches 600 kPa. Determine (a) the mass of the refrigerant in the tank and (b) the amount of heat transferred. Also, show the process on a P-v diagram with respect to saturation lines.

6) Steam at 95 kPa and 5 percent quality is contained in a spring-loaded piston–cylinder device, with an initial volume of 1.2 m^3 . Steam is now heated until its volume is 5.5 m^3 and its pressure is 230 kPa. Determine the heat transferred to and the work produced by the steam during this process.

7) Long cylindrical steel rods ($\rho = 7833 \text{ kg/m}^3$ and $c_p = 0.465 \text{ kJ/kg}\cdot^\circ\text{C}$) of 10 cm diameter are heat-treated by drawing them at a velocity of 3 m/min through an oven maintained at 1000°C . If the rods enter the oven at 30°C and leave at a mean temperature of 400°C , determine the rate of heat transfer to the rods in the oven.

8) An ordinary egg can be approximated as a 5.5-cm-diameter sphere. The egg is initially at a uniform temperature of 4°C and is dropped into boiling water at 97°C . Taking the properties of the egg to be $\rho = 1020 \text{ kg/m}^3$ and $c_p = 3.32 \text{ kJ/kg}\cdot^\circ\text{C}$, determine how much heat is transferred to the egg by the time the average temperature of the egg rises to 90°C .

Enthalpy and Internal Energy

9) A frictionless piston is used to provide a constant pressure of 400 kPa in a cylinder containing steam originally at 200°C with a volume of 2 m^3 . Calculate the final temperature if 3500 kJ of heat is added using the concept of enthalpy and the steam tables.

10) The specific heat of superheated steam at approximately 150 kPa can be determined by the equation below. (a) Determine the enthalpy change between 400°C and 800°C for 4 kg of steam. (b) Determine the average value of C_p between 400°C and 800°C .

$$C_p = 2.07 + \frac{T - 400}{1480} \text{ kJ/kg}\cdot^\circ\text{C}$$

11) Determine the enthalpy change for 2 kg of nitrogen which is heated from 250 K to 1300 K by (a) using the gas tables, (b) assuming constant specific heat. Use $M = 28 \text{ kg/kmol}$.

12) What is the change in the enthalpy, in kJ/kg, of oxygen as its temperature changes from 300 K to 400 K? Is there any difference if the temperature change were from 900 K to 1000 K?